

Q6 Ans: B

Area under the graph gives the elastic potential energy of the spring. As extension is doubled from x_1 to x_2 , area changes from $\frac{1}{2} F_1 x_1$ to $\frac{1}{2} F_2 x_2$. Hence change in EPE = $\frac{1}{2} F_2 x_2 - \frac{1}{2} F_1 x_1$

Q7 Ans: B